WEEK 5 REPORT

Brain MRI Tumour Localisation Using Classical Image Processing

Digital Image Processing | BraTS2020 | BraTS20_Training_001 | Axial Slice 67

1. Objective

The goal was to detect and localise tumour areas within the slice of brain MRI by classical image processing approaches, measure the detected areas, and compare the mask prediction to the segmentation mask.

2. Dataset and Input

The setup involved FLAIR sequence from BraTS20_Training_001 patient at axial slice 67. The segmentation volume was used as the reference for the experiment. Whole tumour means all non-zero segmentation values (seg > 0).

3. Methodology

1. Normalisation and filtering: The MRI image was min-max normalised and filtered with median filter with the kernel size 3×3.
2. Brain region mask: The mask was obtained through applying the Otsu algorithm and some morphological operators. Hole filling and extraction of the largest connected component allowed eliminating the background area.
3. Tumour candidate detection: Otsu thresholding was performed on the intensities within the brain area. The bright areas obtained through this process are candidates but not confirmed tumours.
4. Cleanup: An opening and closing operation was performed using disk size 2, and components below 50 pixels were removed.
5. Localization: The area, centroid, and bounding box of each component were determined using connected component analysis.
6. Evaluation: The predictions were evaluated against the ground-truth mask using Dice, IoU, precision, recall, true positives, false positives, and false negatives.

4. Visual Results

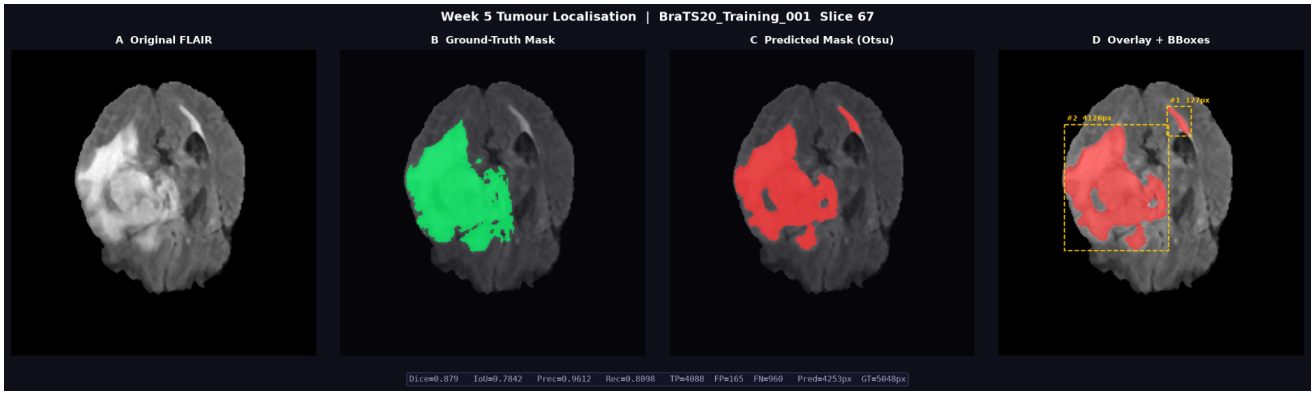


Figure 1. Original FLAIR slice, ground-truth mask, Otsu-predicted mask, and overlay with component bounding boxes.

5. Quantitative Results

Metric	Value
Dice score	0.879
Intersection over Union (IoU)	0.7842
Precision	0.9612
Recall	0.8098
True Positives (Pixels)	4088
False Positives (Pixels)	165
False Negatives (Pixels)	960
Prediction Area (Pixels)	4253
Ground-Truth Area (Pixels)	5048
Otsu's Threshold	0.450232

The mask obtained by prediction has 4253 pixels while the ground truth mask has 5048 pixels. The Dice coefficient is 0.879 and IoU is 0.7842 for this particular slice. These numbers are only applicable to this particular case and not the whole dataset.

6. Connected-Component Localisation

Component	Area	Centroid	Bounding Box (Min row, Min col, Max row, Max col)
1	127	(55.24, 141.91)	(44, 132, 67, 151)
2	4126	(109.71, 89.59)	(58, 51, 157, 133)

The cleaning process produced two connected components of 127 pixels and 4126 pixels. Component labelling refers to the description of related regions in the mask but does not independently confirm whether the region is a tumor region or not.

7. Interpretation

A precision of 0.9612 means that the majority of pixels chosen by the method are indeed tumour pixels in the reference mask. The recall of 0.8098 indicates that there was some amount of true tumour pixels that was not identified. In this case, 960 pixels were left unaccounted, and the predicted region had more pixels than the ground truth region.

8. Limitations

- The main drawback of otsu thresholding is its ability to split the image based on brightness, including bright non-tumorous regions or excluding less-bright tumor regions.
- This report compares a single patient and a single slice, and is not a performance statement for all patients or slices.
- The identified components given by the program are not the tumour region boundaries identified clinically.
- Centroid and Bounding box coordinates are further shown as 2D coordinates rather than in anatomical 3D localization.

9. Conclusion

The classical image processing chain of normalisation, median filtering, Otsu thresholding, morphology cleaning and connected component analysis was applied. The dice score and IoU score of the predicted mask for the slice of BraTS20_Training_001 (axial slice 67) compared to the ground truth are 0.879 and 0.7842 respectively. The experiment has shown how regions can be found in Digital Image Processing.

Appendix: Output Files

- Week5_tumor_localization.png - four-panel visualisation.
- Week5_components.csv - area, center and bounding box coordinates of each component.
- Week5_evaluation.csv - metrics for the chosen slice.
- Week5_summary.md - brief experiment summary.